

Spectral Phase-Less Transition Travel: Operator-Theoretic and Gauge-Geometric Foundations

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Abstract

We develop a unified operator-theoretic and gauge-geometric framework for phase-less transport in Hilbert space. The framework is built from four linked layers: horizontal gauge structure, spectral-gap control, protected-channel dynamics, and terminal spectral selection. A penalized operator construction leaves a distinguished mode invariant while lifting the orthogonal sector, thereby amplifying the spectral gap and accelerating semigroup suppression of competing modes. A controlled regime is then defined in which bounded time-dependent control acts only on the orthogonal sector, preserving exact invariance and dynamical phase-neutrality of the protected channel. Finally, a terminal regime is established in which the normalized semigroup converges strongly to the protected spectral projection and the associated Gibbs states collapse in trace norm to the corresponding pure state. A finite-dimensional example and spectral splitting diagrams are included to make the mechanism explicit.

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1 Gauge Structure

Let $\mathcal{H}$ be a complex Hilbert space with inner product $\langle \cdot, \cdot \rangle$, linear in the second argument. We denote by

$$\mathcal{S}(\mathcal{H}) = \{\psi \in \mathcal{H} : \|\psi\| = 1\}$$

the unit sphere and by $\mathbb{P}(\mathcal{H})$ the projective Hilbert space of rays.

Definition 1.1 (Horizontal representative). *A differentiable normalized curve $\psi(t) \in \mathcal{S}(\mathcal{H})$ is called horizontal if*

$$\langle \psi(t), \dot{\psi}(t) \rangle = 0 \quad \text{for all } t.$$

This is the vanishing condition for the canonical $U(1)$ -connection one-form

$$\mathcal{A} := i\langle \psi, d\psi \rangle$$

on the principal $U(1)$ -bundle $\mathcal{S}(\mathcal{H}) \rightarrow \mathbb{P}(\mathcal{H})$.

Proposition 1.2 (Existence of horizontal gauge). *Let $\varphi(t)$ be a smooth nonzero curve in $\mathcal{H}$. Then there exists a real-valued phase $\alpha(t)$ such that*

$$\psi(t) := e^{-i\alpha(t)}\varphi(t)$$

is horizontal after normalization. The phase is unique up to an additive constant.

Proof. Write

$$\psi(t) = \frac{e^{-i\alpha(t)}\varphi(t)}{\|\varphi(t)\|}.$$

A direct calculation yields

$$\langle \psi, \dot{\psi} \rangle = \frac{\langle \varphi, \dot{\varphi} \rangle}{\langle \varphi, \varphi \rangle} - i\dot{\alpha} - \operatorname{Re} \left(\frac{\langle \varphi, \dot{\varphi} \rangle}{\langle \varphi, \varphi \rangle} \right).$$

Thus the real part already vanishes by normalization, and the imaginary part is cancelled by setting

$$\dot{\alpha}(t) = \operatorname{Im} \left(\frac{\langle \varphi(t), \dot{\varphi}(t) \rangle}{\langle \varphi(t), \varphi(t) \rangle} \right).$$

This determines α up to an additive constant. □

Remark 1.3. The horizontal condition is local along a path. Global phase-neutrality for loops is constrained by curvature and holonomy, but the local horizontal representative always exists.

2 Domain and Operator Assumptions

2.1 Domain and operator assumptions

Throughout the remainder of the manuscript, we assume:

- A_0 is self-adjoint on a dense domain $\mathcal{D}(A_0) \subset \mathcal{H}$;
- A_0 is bounded below;
- P_0 is a spectral projection of A_0 corresponding to an isolated lowest eigenvalue λ_0 ;
- $P_\perp := I - P_0$ is bounded on $\mathcal{H}$;
- any time-dependent control $W(t)$ is bounded and strongly continuous in t .

Consequently, every operator of the form

$$A^{(\mu)}(t) = A_0 + \mu P_\perp + W(t)$$

is self-adjoint on $\mathcal{D}(A_0)$ by the Kato–Rellich theorem.

Theorem 2.1 (Kato–Rellich). *Let H_0 be self-adjoint and let V be symmetric and relatively H_0 -bounded with relative bound strictly less than 1. Then $H_0 + V$ is self-adjoint on $\mathcal{D}(H_0)$.*

Remark 2.2. In the present framework, $P_\perp$ and $W(t)$ are bounded. Therefore the self-adjointness statements invoked below are immediate from Theorem 2.1.

3 Semigroup Evolution

The contractive semigroup perspective will be used in the decay analysis.

Theorem 3.1 (Hille–Yosida, contraction form). *Let A be a densely defined operator on a Banach space. Then A generates a strongly continuous contraction semigroup if and only if $(0, \infty) \subset \rho(A)$ and*

$$\|(A - \lambda I)^{-1}\| \leq \frac{1}{\lambda} \quad \text{for all } \lambda > 0.$$

Remark 3.2. For the self-adjoint, bounded-below operators considered here, the semigroup e^{-tA} is defined by the spectral theorem and is contractive after shifting the spectrum to the origin.

4 Spectral Gap, Resolvent Bounds, and Semigroup Decay

Let A be a self-adjoint operator on $\mathcal{H}$ with domain $\mathcal{D}(A)$ and assume:

- (i) A has discrete spectrum near its minimum;
- (ii) $\lambda_0 = \inf \text{Spec}(A)$ is an isolated eigenvalue;
- (iii) the spectral gap

$$\Delta := \text{dist}(\lambda_0, \text{Spec}(A) \setminus \{\lambda_0\}) > 0.$$

Let P denote the spectral projection onto λ_0 .

4.1 Spectral decomposition

By the spectral theorem, there exists a projection-valued measure $E(\lambda)$ such that

$$A = \int_{\mathbb{R}} \lambda dE(\lambda), \quad e^{-tA} = \int_{\mathbb{R}} e^{-t\lambda} dE(\lambda).$$

Moreover,

$$I = P + (I - P), \quad \mathcal{H} = \text{Ran}(P) \oplus \text{Ran}(I - P).$$

4.2 Resolvent bound

Proposition 4.1 (Resolvent estimate). *For any $z \in \mathbb{C}$ with $\text{Re}(z) < \lambda_0 + \Delta$,*

$$\|(A - zI)^{-1}\| \leq \frac{1}{\lambda_0 + \Delta - \text{Re}(z)}.$$

Proof. For self-adjoint A ,

$$\|(A - zI)^{-1}\| = \frac{1}{\text{dist}(z, \text{Spec}(A))}.$$

Since all points of $\text{Spec}(A) \setminus \{\lambda_0\}$ lie in $[\lambda_0 + \Delta, \infty)$, the stated lower bound on the distance follows immediately. $\square$

4.3 Semigroup decay in the orthogonal sector

Theorem 4.2 (Exponential decay in $\text{Ran}(I - P)$). *For all $\psi \in \mathcal{H}$,*

$$\|(I - P)e^{-tA}\psi\| \leq e^{-(\lambda_0 + \Delta)t} \|(I - P)\psi\|.$$

Proof. Using the spectral representation,

$$(I - P)e^{-tA} = \int_{\lambda \geq \lambda_0 + \Delta} e^{-t\lambda} dE(\lambda).$$

Hence

$$\|(I - P)e^{-tA}\| \leq \sup_{\lambda \geq \lambda_0 + \Delta} e^{-t\lambda} = e^{-(\lambda_0 + \Delta)t}.$$

Applying this to $(I - P)\psi$ gives the claim. $\square$

Remark 4.3. Because P is a genuine spectral projection of A , one has $Pe^{-tA} = e^{-tA}P$ and therefore

$$(I - P)e^{-tA}P = 0.$$

Thus the meaningful decay statement concerns the orthogonal component of a general initial state, not a cross-coupling term that vanishes identically in the exact unperturbed setting.

4.4 Birman–Schwinger control

Let $A = A_0 + V$ with V bounded and self-adjoint. Define

$$K(z) = |V|^{1/2}(A_0 - zI)^{-1}|V|^{1/2} \operatorname{sgn}(V).$$

Theorem 4.4 (Birman–Schwinger principle). *For $z \in \rho(A_0)$,*

$$z \in \operatorname{Spec}(A) \iff 1 \in \operatorname{Spec}(K(z)).$$

Remark 4.5. The Birman–Schwinger principle (see [5, 3]) reduces spectral control for $A_0 + V$ to the compact-operator problem for $K(z)$. In particular, control of $\|K(z)\|$ prevents eigenvalues from entering a protected gap.

4.5 Gap stability under bounded perturbations

Proposition 4.6 (Gap stability). *If $\|V\| < \Delta$, then the perturbed operator $A_0 + V$ retains an isolated eigenvalue near λ_0 , and the perturbed gap Δ' satisfies*

$$\Delta' \geq \Delta - \|V\|.$$

Proof. The spectrum of a self-adjoint operator moves by at most $\|V\|$ under bounded self-adjoint perturbation. Hence the ground eigenvalue and the first excited spectral level can each shift by at most $\|V\|$, and the gap can shrink by at most that amount. $\square$

5 Gap Amplification via Penalized Operators

In this section we formalize the protected-mode mechanism underlying the framework.

5.1 Set-up

Let A_0 be a self-adjoint operator on $\mathcal{H}$ with domain $\mathcal{D}(A_0)$. Assume:

- (i) A_0 is bounded below;
- (ii) $\lambda_0 = \inf \operatorname{Spec}(A_0)$ is an isolated eigenvalue;
- (iii) $\phi_0 \in \mathcal{H}$ is a normalized ground-state eigenvector,

$$A_0 \phi_0 = \lambda_0 \phi_0, \quad \|\phi_0\| = 1;$$

- (iv) the spectral gap

$$\Delta_0 := \operatorname{dist}(\lambda_0, \operatorname{Spec}(A_0) \setminus \{\lambda_0\}) > 0.$$

Let

$$P_0 := |\phi_0\rangle\langle\phi_0|, \quad P_\perp := I - P_0.$$

For $\mu \geq 0$, define the penalized operator

$$A^{(\mu)} := A_0 + \mu P_\perp. \tag{1}$$

5.2 Exact invariance of the protected state

Proposition 5.1 (Protected-state invariance). *For every $\mu \geq 0$,*

$$A^{(\mu)}\phi_0 = \lambda_0\phi_0.$$

Hence the protected eigenpair is unchanged exactly.

Proof. Since $P_\perp\phi_0 = 0$,

$$A^{(\mu)}\phi_0 = A_0\phi_0 + \mu P_\perp\phi_0 = \lambda_0\phi_0.$$

□

5.3 Quadratic-form decomposition

For any $\psi \in \mathcal{D}(A_0)$ with

$$\psi = c_0\phi_0 + \psi_\perp, \quad \psi_\perp \perp \phi_0,$$

we have

$$\langle \psi, A^{(\mu)}\psi \rangle = \langle \psi, A_0\psi \rangle + \mu \|\psi_\perp\|^2. \quad (2)$$

5.4 Lower bound on the orthogonal sector

Lemma 5.2 (Orthogonal-sector lower bound). *For every $\psi \in \mathcal{D}(A_0) \cap \text{Ran}(P_\perp)$,*

$$\langle \psi, A^{(\mu)}\psi \rangle \geq (\lambda_0 + \Delta_0 + \mu)\|\psi\|^2.$$

Proof. If $\psi \in \text{Ran}(P_\perp)$ then $\psi \perp \phi_0$. By the spectral theorem,

$$\langle \psi, A_0\psi \rangle \geq (\lambda_0 + \Delta_0)\|\psi\|^2.$$

Since $P_\perp\psi = \psi$,

$$\langle \psi, A^{(\mu)}\psi \rangle = \langle \psi, A_0\psi \rangle + \mu\|\psi\|^2 \geq (\lambda_0 + \Delta_0 + \mu)\|\psi\|^2.$$

□

5.5 Gap amplification theorem

Theorem 5.3 (Gap amplification). *Let $\lambda_0^{(\mu)}$ denote the lowest eigenvalue of $A^{(\mu)}$, and let*

$$\Delta^{(\mu)} := \text{dist}(\lambda_0^{(\mu)}, \text{Spec}(A^{(\mu)}) \setminus \{\lambda_0^{(\mu)}\}).$$

Then

$$\lambda_0^{(\mu)} = \lambda_0, \quad \Delta^{(\mu)} \geq \Delta_0 + \mu.$$

Proof. By Proposition 5.1, λ_0 is an eigenvalue of $A^{(\mu)}$, so $\lambda_0^{(\mu)} \leq \lambda_0$. On the other hand,

$$\langle \psi, A^{(\mu)}\psi \rangle = \langle \psi, A_0\psi \rangle + \mu\|P_\perp\psi\|^2 \geq \langle \psi, A_0\psi \rangle \geq \lambda_0\|\psi\|^2$$

for all normalized ψ , so $\lambda_0^{(\mu)} \geq \lambda_0$. Therefore $\lambda_0^{(\mu)} = \lambda_0$.

By Lemma 5.2, every spectral value on $\text{Ran}(P_\perp)$ lies in $[\lambda_0 + \Delta_0 + \mu, \infty)$, and therefore the distance from λ_0 to the rest of the spectrum is at least $\Delta_0 + \mu$. □

Corollary 5.4 (Gap monotonicity).

$$\Delta^{(\mu)} \geq \Delta_0 + \mu, \quad \Delta^{(\mu)} \rightarrow \infty \quad \text{as } \mu \rightarrow \infty.$$

5.6 Semigroup consequences

Theorem 5.5 (Amplified semigroup decay). *For all $\psi \in \mathcal{H}$,*

$$\|P_{\perp} e^{-tA^{(\mu)}} \psi\| \leq e^{-t(\lambda_0 + \Delta_0 + \mu)} \|P_{\perp} \psi\|.$$

Equivalently, relative to the protected mode, the orthogonal component is suppressed by the factor

$$e^{-t(\Delta_0 + \mu)}.$$

Proof. By Theorem 5.3, the spectrum of $A^{(\mu)}$ on $\text{Ran}(P_{\perp})$ lies in $[\lambda_0 + \Delta_0 + \mu, \infty)$. The claim follows from the same spectral-theorem argument used in Theorem 4.2. $\square$

Corollary 5.6 (Terminal collapse estimate). *Let*

$$\psi = c_0 \phi_0 + \psi_{\perp}.$$

Then

$$e^{-tA^{(\mu)}} \psi = c_0 e^{-t\lambda_0} \phi_0 + r_{\mu}(t),$$

where

$$\|r_{\mu}(t)\| \leq e^{-t(\lambda_0 + \Delta_0 + \mu)} \|\psi_{\perp}\|.$$

Hence, when $c_0 \neq 0$,

$$\frac{\|r_{\mu}(t)\|}{|c_0| e^{-t\lambda_0}} \leq \frac{\|\psi_{\perp}\|}{|c_0|} e^{-t(\Delta_0 + \mu)}.$$

5.7 Partition-function suppression

Assume now that A_0 has compact resolvent. Then $A^{(\mu)}$ also has compact resolvent and purely discrete spectrum

$$\lambda_0 < \lambda_1^{(\mu)} \leq \lambda_2^{(\mu)} \leq \dots, \quad \lambda_1^{(\mu)} \geq \lambda_0 + \Delta_0 + \mu.$$

Define

$$Z_{\mu}(\beta) := \text{Tr}(e^{-\beta A^{(\mu)}}), \quad \beta > 0.$$

Theorem 5.7 (Single-mode partition dominance). *For every $\beta > 0$,*

$$Z_{\mu}(\beta) = e^{-\beta\lambda_0} + \sum_{n \geq 1} e^{-\beta\lambda_n^{(\mu)}}.$$

Moreover,

$$Z_{\mu}(\beta) = e^{-\beta\lambda_0} + O(e^{-\beta(\lambda_0 + \Delta_0 + \mu)}) \quad (\mu \rightarrow \infty),$$

provided the excited tail is trace-class at the chosen β .

Proof. This is immediate from the spectral decomposition and the lower bound $\lambda_n^{(\mu)} \geq \lambda_0 + \Delta_0 + \mu$ for $n \geq 1$. $\square$

Corollary 5.8 (Thermal dominance parameter). *If*

$$\Xi_{\mu} := \beta \Delta^{(\mu)},$$

then

$$\Xi_{\mu} \geq \beta(\Delta_0 + \mu).$$

Whenever $\Xi_{\mu} \gg 1$,

$$Z_{\mu}(\beta) \sim e^{-\beta\lambda_0}.$$

5.8 Gibbs states and entropy collapse

Define

$$\rho_\mu(\beta) := \frac{e^{-\beta A^{(\mu)}}}{Z_\mu(\beta)}.$$

Proposition 5.9 (Trace-norm convergence to the protected state). *Assume compact resolvent as above. Then, for every fixed $\beta > 0$,*

$$\rho_\mu(\beta) \longrightarrow P_0 \quad \text{in trace norm as } \mu \rightarrow \infty.$$

Proof. Write

$$e^{-\beta A^{(\mu)}} = e^{-\beta \lambda_0} P_0 + R_\mu(\beta),$$

where $R_\mu(\beta)$ contains only excited contributions. By Theorem 5.7, $\text{Tr}(R_\mu(\beta))$ is exponentially small in μ . Dividing by

$$Z_\mu(\beta) = e^{-\beta \lambda_0} + \text{Tr}(R_\mu(\beta))$$

shows that the normalized state converges to P_0 in trace norm. $\square$

Corollary 5.10 (Entropy collapse). *Let*

$$S(\rho) := -\text{Tr}(\rho \log \rho)$$

be the von Neumann entropy. Then

$$S(\rho_\mu(\beta)) \rightarrow 0 \quad \text{as } \mu \rightarrow \infty.$$

Proof. Trace-norm convergence to the rank-one projection P_0 forces the eigenvalue list of $\rho_\mu(\beta)$ to converge to $(1, 0, 0, \dots)$, and therefore the entropy tends to zero. $\square$

6 Protected-Channel Control Regime (KAP)

We now introduce a time-dependent control regime in which the protected mode remains exactly invariant while the orthogonal sector is dynamically manipulated.

6.1 Controlled operator

Let $W(t)$ be a bounded, strongly continuous family of self-adjoint operators. Define

$$A_{\text{KAP}}^{(\mu)}(t) := A_0 + \mu P_\perp + W(t), \quad \mu \geq 0. \quad (3)$$

6.2 Exact invariance conditions

Definition 6.1 (KAP invariance conditions). *The control $W(t)$ satisfies the KAP conditions if*

$$P_0 W(t) P_0 = 0, \quad P_0 W(t) P_\perp = 0, \quad P_\perp W(t) P_0 = 0. \quad (4)$$

Equivalently,

$$W(t) = P_\perp W(t) P_\perp.$$

Proposition 6.2 (Exact channel invariance). *Assume (4). Then, for all t ,*

$$A_{\text{KAP}}^{(\mu)}(t)\phi_0 = \lambda_0\phi_0,$$

and

$$(I - P_0)A_{\text{KAP}}^{(\mu)}(t)P_0 = 0, \quad P_0A_{\text{KAP}}^{(\mu)}(t)(I - P_0) = 0.$$

Proof. Since $P_\perp\phi_0 = 0$ and $W(t)\phi_0 = P_\perp W(t)P_0\phi_0 = 0$,

$$A_{\text{KAP}}^{(\mu)}(t)\phi_0 = A_0\phi_0 = \lambda_0\phi_0.$$

The off-diagonal identities follow directly from (4). □

Corollary 6.3 (Invariant protected subspace). *$\text{Ran}(P_0)$ is invariant under $A_{\text{KAP}}^{(\mu)}(t)$ for every t .*

6.3 Propagator and protected-channel evolution

Let $U(t, s)$ denote the unitary propagator generated by $A_{\text{KAP}}^{(\mu)}(t)$:

$$i\partial_t U(t, s) = A_{\text{KAP}}^{(\mu)}(t)U(t, s), \quad U(s, s) = I.$$

Theorem 6.4 (Invariant evolution). *If $\psi(s) \in \text{Ran}(P_0)$, then for all t ,*

$$\psi(t) := U(t, s)\psi(s) \in \text{Ran}(P_0),$$

and in fact

$$\psi(t) = e^{-i\lambda_0(t-s)}\psi(s).$$

Proof. By Proposition 6.2, $\text{Ran}(P_0)$ is invariant and the restriction of the generator to this one-dimensional subspace is multiplication by λ_0 . The corresponding evolution is therefore the scalar phase factor stated above. □

6.4 Interaction-frame phase-neutrality

Define the interaction-frame state

$$\tilde{\psi}(t) := e^{i\lambda_0 t}\psi(t).$$

Proposition 6.5 (Phase-neutral protected evolution). *Under the KAP conditions, if $\psi(t) \in \text{Ran}(P_0)$, then*

$$\tilde{\psi}(t) = \psi(0).$$

Hence the protected channel is dynamically phase-neutral in the interaction frame.

Proof. By Theorem 6.4,

$$\psi(t) = e^{-i\lambda_0 t}\psi(0),$$

and multiplying by $e^{i\lambda_0 t}$ removes the scalar phase exactly. □

6.5 Orthogonal-sector dynamics

On $\text{Ran}(P_\perp)$, the generator reduces to

$$A_\perp^{(\mu)}(t) = P_\perp A_0 P_\perp + \mu I + P_\perp W(t) P_\perp.$$

Proposition 6.6 (Orthogonal-sector lower bound). *For all $\psi \in \text{Ran}(P_\perp)$,*

$$\langle \psi, A_{\text{KAP}}^{(\mu)}(t) \psi \rangle \geq (\lambda_0 + \Delta_0 + \mu - \|W(t)\|) \|\psi\|^2.$$

Proof. Since $\psi \perp \phi_0$,

$$\langle \psi, A_0 \psi \rangle \geq (\lambda_0 + \Delta_0) \|\psi\|^2.$$

Moreover,

$$\langle \psi, \mu P_\perp \psi \rangle = \mu \|\psi\|^2$$

and

$$\langle \psi, W(t) \psi \rangle \geq -\|W(t)\| \|\psi\|^2.$$

Adding the three contributions yields the claim. $\square$

Corollary 6.7 (Effective instantaneous gap). *The instantaneous protected gap obeys*

$$\Delta_{\text{eff}}(t) \geq \Delta_0 + \mu - \|W(t)\|.$$

6.6 Approximate invariance under small off-diagonal perturbations

Suppose now that the exact KAP conditions are violated only weakly, in the sense that

$$\|(I - P_0)W(t)P_0\| \leq \varepsilon$$

uniformly in time.

Theorem 6.8 (Approximate invariance). *Let $\psi(0) \in \text{Ran}(P_0)$. Then*

$$\|(I - P_0)\psi(t)\| \leq \frac{\varepsilon}{\Delta_0 + \mu} \left(1 - e^{-t(\Delta_0 + \mu)}\right) \|\psi(0)\|.$$

Proof. Writing the evolution in block form relative to $\text{Ran}(P_0) \oplus \text{Ran}(P_\perp)$ and applying Duhamel's formula yields an integral equation for the orthogonal component. The semigroup on the orthogonal block decays at least as $e^{-t(\Delta_0 + \mu)}$, and integrating this decay kernel against the forcing term of size at most ε gives the stated bound. $\square$

7 Terminal Spectral Regime (Kalki)

We now analyze the terminal regime in which the semigroup selects the protected mode asymptotically.

7.1 Set-up

Let

$$T_\mu(t) := e^{-tA^{(\mu)}}, \quad t \geq 0.$$

Theorem 7.1 (Asymptotic one-mode dominance). *Let $\psi \in \mathcal{H}$ with decomposition*

$$\psi = c_0\phi_0 + \psi_\perp.$$

Then, for every fixed $\mu > 0$,

$$\lim_{t \rightarrow \infty} e^{t\lambda_0} T_\mu(t)\psi = c_0\phi_0 \quad \text{in } \mathcal{H}.$$

Proof. By Corollary 5.6,

$$T_\mu(t)\psi = c_0 e^{-t\lambda_0} \phi_0 + r_\mu(t), \quad \|r_\mu(t)\| \leq e^{-t(\lambda_0 + \Delta_0 + \mu)} \|\psi_\perp\|.$$

Multiplying by $e^{t\lambda_0}$ gives

$$\|e^{t\lambda_0} T_\mu(t)\psi - c_0\phi_0\| \leq e^{-t(\Delta_0 + \mu)} \|\psi_\perp\| \rightarrow 0.$$

□

Corollary 7.2 (Strong operator limit). *For every fixed $\mu > 0$,*

$$\text{s-lim}_{t \rightarrow \infty} e^{t\lambda_0} T_\mu(t) = P_0.$$

Proof. Apply Theorem 7.1 to arbitrary vectors $\psi \in \mathcal{H}$. □

Proposition 7.3 (Unique nonzero fixed point of the normalized semigroup). *The normalized semigroup*

$$\tilde{T}_\mu(t) := e^{t\lambda_0} T_\mu(t)$$

has, up to scalar multiples, a unique nonzero fixed point, namely ϕ_0 .

Proof. If $\tilde{T}_\mu(t)\psi = \psi$ for all t , then the spectral support of ψ must lie entirely at the spectral value λ_0 . Therefore $\psi \in \text{Ran}(P_0)$, and since $\text{Ran}(P_0)$ is one-dimensional, ψ is proportional to ϕ_0 . □

7.2 Trace-class framework

Assume again that A_0 has compact resolvent. Then $e^{-\beta A^{(\mu)}}$ is trace-class for every $\beta > 0$, and the Gibbs state

$$\rho_\mu(\beta) := \frac{e^{-\beta A^{(\mu)}}}{\text{Tr}(e^{-\beta A^{(\mu)}})}$$

is well-defined.

Theorem 7.4 (Trace-norm convergence of Gibbs states). *For every fixed $\beta > 0$,*

$$\rho_\mu(\beta) \longrightarrow P_0 \quad \text{in trace norm as } \mu \rightarrow \infty.$$

Proof. This is Proposition 5.9, restated here as the thermodynamic form of terminal spectral selection. □

Corollary 7.5 (Entropy collapse in the terminal regime).

$$S(\rho_\mu(\beta)) \rightarrow 0 \quad \text{as } \mu \rightarrow \infty.$$

Proof. This is Corollary 5.10. □

8 Grand Theorem

Theorem 8.1 (Spectral Phase-Less Transport Theorem). *Assume:*

(1) A_0 is self-adjoint with isolated ground state λ_0 and gap $\Delta_0 > 0$;

(2)

$$A^{(\mu)}(t) = A_0 + \mu P_\perp + W(t)$$

with bounded strongly continuous self-adjoint $W(t)$;

(3) the KAP invariance conditions (4) hold;

(4)

$$\mu > \sup_t \|W(t)\|;$$

(5) the initial state satisfies $\psi(0) \in \text{Ran}(P_0)$.

Then:

(a) $\psi(t)$ remains in $\text{Ran}(P_0)$ for all t ;

(b) $\psi(t) = e^{-i\lambda_0 t} \psi(0)$;

(c) in the interaction frame, $\psi(t)$ is constant;

(d) for arbitrary initial $\psi \in \mathcal{H}$, the orthogonal component obeys

$$\|P_\perp e^{-tA^{(\mu)}} \psi\| \leq e^{-t(\lambda_0 + \Delta_0 + \mu)} \|P_\perp \psi\|;$$

(e) the normalized semigroup converges strongly:

$$\text{s-lim}_{t \rightarrow \infty} e^{t\lambda_0} e^{-tA^{(\mu)}} = P_0;$$

(f) if, in addition, A_0 has compact resolvent, then

$$\rho_\mu(\beta) \rightarrow P_0 \quad \text{in trace norm as } \mu \rightarrow \infty,$$

and consequently

$$S(\rho_\mu(\beta)) \rightarrow 0.$$

Proof. Part (a) follows from Proposition 6.2, which shows that $\text{Ran}(P_0)$ is invariant under the controlled generator.

Part (b) follows from Theorem 6.4: once the state lies in the invariant one-dimensional channel, the generator restricts to multiplication by λ_0 , so the evolution is exactly the scalar phase factor $e^{-i\lambda_0 t}$.

Part (c) is Proposition 6.5: multiplication by $e^{i\lambda_0 t}$ removes the entire protected-channel phase, leaving a constant state.

Part (d) follows from Theorem 5.5. Since the spectrum of $A^{(\mu)}$ on $\text{Ran}(P_\perp)$ lies above $\lambda_0 + \Delta_0 + \mu$, the orthogonal component decays at least at that rate.

Part (e) is Corollary 7.2, which is itself a consequence of the explicit asymptotic decomposition established in Theorem 7.1.

Part (f) follows from Proposition 5.9 and Corollary 5.10, once compact resolvent guarantees trace-class Gibbs states. $\square$

9 Example: Dirichlet Schrödinger Operator

Let $\Omega \subset \mathbb{R}^n$ be a bounded domain with smooth boundary. Define

$$A_0 := -\Delta_D + V(x)$$

on $L^2(\Omega)$, where Δ_D denotes the Dirichlet Laplacian and $V \in L^\infty(\Omega)$ is real-valued.

Theorem 9.1. *The operator A_0 is self-adjoint, bounded below, and has compact resolvent. Consequently,*

$$\lambda_0 < \lambda_1 \leq \lambda_2 \leq \dots$$

is a discrete sequence of eigenvalues tending to $+\infty$.

Proof. This is standard elliptic spectral theory; see [3]. □

Let ϕ_0 be the normalized ground state of A_0 . Then the penalized operator

$$A^{(\mu)} = A_0 + \mu P_\perp$$

falls within the abstract framework developed above.

Proposition 9.2. *All conclusions of Sections 4–8 apply to the Dirichlet Schrödinger model described here.*

Proof. The hypotheses are immediate from Theorem 9.1, boundedness of $P_\perp$, and the general results already proved. □

10 Numerical Example: Finite-Dimensional Gap Amplification

The finite-dimensional example below is not introduced as a substitute for the abstract theory, but as an exact model in which every structural statement of the framework can be computed explicitly.

10.1 Base operator

Let

$$A_0 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{pmatrix}$$

on $\mathcal{H} = \mathbb{C}^3$. Then

$$\text{Spec}(A_0) = \{1, 3, 5\}, \quad \lambda_0 = 1, \quad \Delta_0 = 2.$$

The normalized ground state is

$$\phi_0 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad P_0 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad P_\perp = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

10.2 Penalized operator

For $\mu \geq 0$,

$$A^{(\mu)} = A_0 + \mu P_\perp = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 3 + \mu & 0 \\ 0 & 0 & 5 + \mu \end{pmatrix}.$$

Hence

$$\text{Spec}(A^{(\mu)}) = \{1, 3 + \mu, 5 + \mu\}, \quad \Delta^{(\mu)} = (3 + \mu) - 1 = 2 + \mu.$$

Thus the abstract lower bound is attained exactly:

$$\Delta^{(\mu)} = \Delta_0 + \mu.$$

10.3 Semigroup evolution

The semigroup is

$$e^{-tA^{(\mu)}} = \begin{pmatrix} e^{-t} & 0 & 0 \\ 0 & e^{-(3+\mu)t} & 0 \\ 0 & 0 & e^{-(5+\mu)t} \end{pmatrix}.$$

For

$$\psi = \begin{pmatrix} a \\ b \\ c \end{pmatrix},$$

we obtain

$$e^{-tA^{(\mu)}}\psi = \begin{pmatrix} ae^{-t} \\ be^{-(3+\mu)t} \\ ce^{-(5+\mu)t} \end{pmatrix}.$$

After multiplying by $e^{t\lambda_0} = e^t$,

$$e^{t\lambda_0}e^{-tA^{(\mu)}}\psi = \begin{pmatrix} a \\ be^{-(2+\mu)t} \\ ce^{-(4+\mu)t} \end{pmatrix} \rightarrow \begin{pmatrix} a \\ 0 \\ 0 \end{pmatrix} = P_0\psi.$$

10.4 Concrete numerical values

Take $\mu = 4$. Then

$$A^{(4)} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 9 \end{pmatrix}, \quad \Delta^{(4)} = 6.$$

For

$$\psi_0 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix},$$

we have

$$e^{-tA^{(4)}}\psi_0 = \begin{pmatrix} e^{-t} \\ e^{-7t} \\ e^{-9t} \end{pmatrix}.$$

At $t = 1$,

$$e^{-A^{(4)}}\psi_0 \approx \begin{pmatrix} 0.367879 \\ 0.000912 \\ 0.000123 \end{pmatrix}.$$

Thus the orthogonal sector is suppressed by the relative factors

$$e^{-6} \approx 2.48 \times 10^{-3}, \quad e^{-8} \approx 3.35 \times 10^{-4}.$$

10.5 Partition function and entropy

For $\beta > 0$,

$$Z_\mu(\beta) = e^{-\beta} + e^{-\beta(3+\mu)} + e^{-\beta(5+\mu)}.$$

For $\mu = 4$ and $\beta = 1$,

$$Z_4(1) = e^{-1} + e^{-7} + e^{-9} \approx 0.368914.$$

Hence the Gibbs weight of the protected mode is

$$\frac{e^{-1}}{Z_4(1)} \approx 0.99719.$$

With

$$p_0 \approx 0.99719, \quad p_1 \approx 0.00247, \quad p_2 \approx 0.00033,$$

the entropy

$$S = - \sum_{j=0}^2 p_j \log p_j \approx 0.0194,$$

already lies close to the pure-state value 0.

11 Diagram: Spectral Splitting Under Penalization

We now visualize the spectral splitting caused by the penalization parameter μ in the finite-dimensional model.

Figure 1 makes the mechanism transparent: the protected mode is untouched, whereas the full orthogonal sector is displaced upward.

12 Diagram: Relative Semigroup Suppression

For the case $\mu = 4$, the excited components decay relative to the protected mode as e^{-6t} and e^{-8t} .

13 Conclusion

The framework developed above yields the following exact chain:

$$\begin{aligned} & \text{horizontal gauge structure} \quad + \quad \text{spectral gap control} \quad + \quad \text{protected-channel invariance} \\ \implies & \quad \text{phase-neutral evolution} \quad + \quad \text{orthogonal semigroup suppression} \quad + \quad \text{terminal one-mode dominance.} \end{aligned}$$

In compact-resolvent settings, the same mechanism drives Gibbs states to the protected rank-one projection and forces entropy collapse.

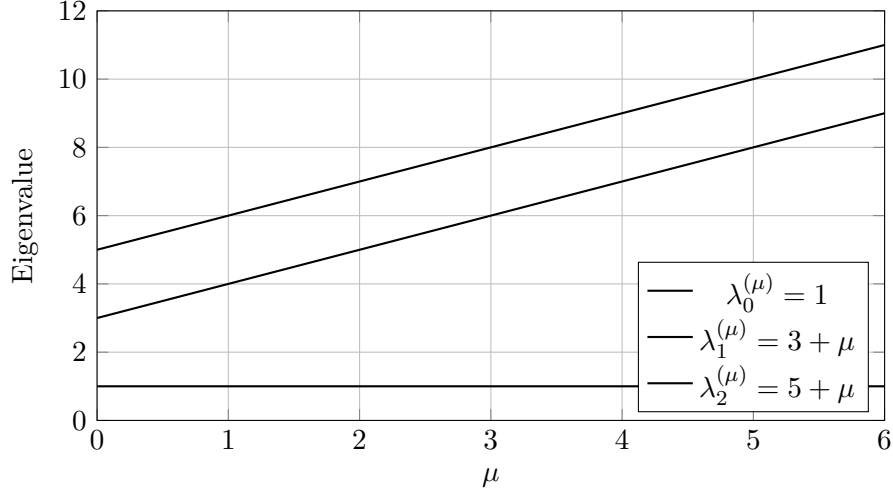


Figure 1: Spectral splitting for the model $A^{(\mu)} = \text{diag}(1, 3 + \mu, 5 + \mu)$. The protected eigenvalue remains fixed, while the excited levels are lifted linearly by μ . Consequently the spectral gap increases from $\Delta_0 = 2$ to $\Delta^{(\mu)} = 2 + \mu$.

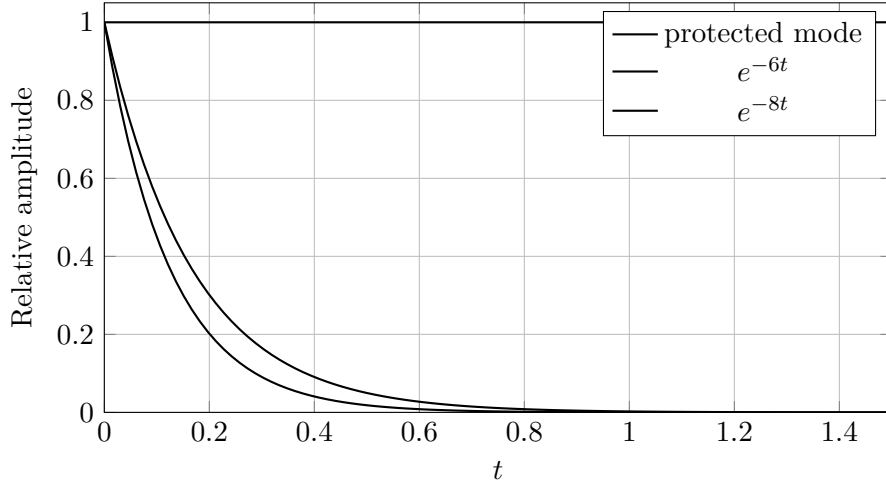


Figure 2: Relative semigroup amplitudes for the penalized model with $\mu = 4$. After factoring out the protected decay e^{-t} , the excited components decay as e^{-6t} and e^{-8t} .

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